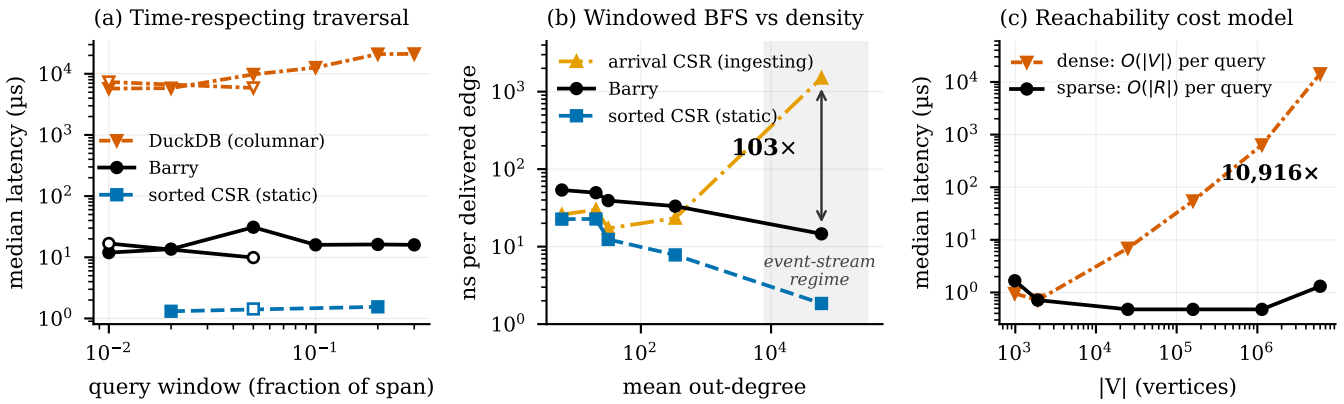
(a) Time-respecting traversal
median latency (µs)
DuckDB (columnar)
Barry
sorted CSR (static)
query window (fraction of span)
(b) Windowed BFS vs density
ns per delivered edge
arrival CSR (ingesting)
Barry
sorted CSR (static)
103×
event-stream regime
mean out-degree
(c) Reachability cost model
median latency (µs)
dense: O(|V|) per query
sparse: O(|R|) per query
10,916×
|V| (vertices)